DOCUMENT C | SOLUTIONS 2

Worked solutions and proposed marking scheme

Alternative predicted practice paper. These are independent suggested answers and mark allocations, not official university solutions.

Award method marks for correct reasoning and units even when a later arithmetic slip occurs; avoid penalizing the same carried error repeatedly. Equivalent physically correct explanations and valid stated assumptions can earn credit. A part's marking points total its question allocation. Diagrams are schematic and not to scale. The complete question text is repeated below so each solution file is usable independently.

Source notation

S1-S06 refer to the supplied six lecture decks and PDF page/slide numbers. A1/A2 are the supplied thermal/OCT student assignments, cited by PDF page including covers. "Recurring family" identifies the historical concept, not copied wording; original extensions are labeled. A historical reference means the closest related question, not evidence that the exact model task appeared.

Equation corrections

Positive activity is -dN/dt. Conventional spin echoes use intrinsic T2 in the ideal echo envelope, whereas gradient echoes retain T2* sensitivity. RF bandwidth means full width unless explicitly stated otherwise. All calculations use γ=267.52×10^6 rad s^-1 T^-1 and unrounded intermediate values.

Model 2 | Question 1: Image acquisition and radiotracer kinetics

1(a) [3 marks]

Question. Name three modalities using ionizing radiation and three non-ionizing modalities. For this answer, treat PET and SPECT as distinct modalities.

Expected answer and working. Ionizing examples: projection X-ray, CT and PET (SPECT also accepted). Non-ionizing examples: MRI, ultrasound and OCT (thermal also accepted). Their source energies/interactions differ; the classification is not whether the system requires external power.

Proposed allocation: Three correct ionizing examples (1.5); Three correct non-ionizing examples (1.5). **Total 3.**

Source map: S1 pp.6,29. **Closest past question:** 2023 Q1(a)i; 2024 Q1(a)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(b) [4 marks]

Question. For CT, MRI, B-mode ultrasound and PET, state the principal physical quantity or signal determining pixel brightness. Include one qualification for MRI or ultrasound.

Expected answer and working. CT estimates linear X-ray attenuation coefficient, commonly expressed as HU relative to water. MRI shows sequence-weighted signal from mobile proton density and relaxation (not simply water concentration). B-mode ultrasound brightness encodes processed echo amplitude/backscatter from tissue interfaces and scatterers, influenced by impedance differences. PET estimates tracer activity distribution after reconstruction and corrections. MRI sequence settings or US gain/processing must be specified for quantitative brightness interpretation.

Proposed allocation: CT attenuation/HU (1); MRI density/relaxation qualification (1); US processed echoes (1); PET activity distribution (1). **Total 4.**

Source map: S1 p.31; S4 p.75; S05 p.19; S06 pp.25-30. **Closest past question:** 2023 Q1(a)ii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(c) [3 marks]

Question. Define quantization error and explain two ways to reduce its relative importance in image acquisition.

Expected answer and working. Quantization error is the difference between the analogue input and the represented quantized value. For an ideal rounding ADC it lies within ±half a code step. Increase bit depth to reduce the code step, or amplify the useful signal to occupy the ADC input range without saturation. Adequately dithered oversampling/filtering is another valid approach.

Proposed allocation: Definition (1); Greater bit depth (1); Use input range without clipping or valid oversampling explanation (1). **Total 3.**

Source map: S1 pp.62-72. **Closest past question:** 2023 Q1(b)ii-iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(d) [4 marks]

Question. A laboratory needs 18 MBq at 10:00 tomorrow. A sealed tracer stock with physical half-life 4 h is purchased at 18:00 today. Neglect biological clearance. Find the required purchase activity.

Expected answer and working. The elapsed time is 16 h, equivalent to four half-lives. A(t)=A0·2^(-t/T), hence A0=A(t)·2^(t/T)=18·2^4=288 MBq. This is the activity at purchase, before the 16 h decay period; using biological half-life would be inappropriate for a sealed stock.

Proposed allocation: 16 h interval (1); Decay law rearranged (1); Four half-lives / factor 16 (1); 288 MBq (1). **Total 4.**

Source map: S06 pp.6-7. **Closest past question:** 2023 Q1(c)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(e) [5 marks]

Question. Initially there are equal numbers of nuclei of tracers X and Y. Their half-lives are 4 h and 20 h. With equal and constant count-detection efficiencies and no biological clearance, find the earliest time at which X contributes only 25% of their combined count rate.

Expected answer and working. Activity is λN, so AX0/AY0=λX/λY=20/4=5, not 1. The condition AX/(AX+AY)=0.25 gives AX/AY=1/3. The ratio evolves as 5 exp[-(λX-λY)t]. Thus exp[-(ln2)(1/4-1/20)t]=1/15 and t=ln15/[ln2·0.20]=19.53445 h. The fraction is below 25% only after this threshold.

Proposed allocation: Initial activity ratio 5 (1); Fraction becomes ratio 1/3 (1); Ratio decay equation (1); Logarithmic solution (1); 19.53 h / threshold interpretation (1). **Total 5.**

Source map: S06 pp.6-7. **Closest past question:** 2025 Q1(b)iv. **Status:** Recurring family.

Typical mark-losing error: Equal initial nuclei do not mean equal initial count rates.

1(f) [6 marks]

Question. Compare SPECT and PET in terms of emission, direction selection and detection sensitivity. Give one practical limitation of PET.

Expected answer and working. SPECT uses single gamma emission, while PET detects the two 511 keV photons following positron annihilation. SPECT obtains direction information with a physical collimator that rejects most photons; PET uses coincidence-defined LORs. PET can collect a larger fraction of useful events with a detector ring and no comparable in-plane mechanical collimator, and its higher photon energy reduces tissue attenuation. Sensitivity and SNR still depend on acquisition design and correction quality. PET requires reliable short-lived tracer supply/production and expensive equipment; an on-site cyclotron is not a universal requirement for every tracer.

Proposed allocation: Single gamma vs annihilation pair (2); Physical collimator vs coincidence LOR (2); Sensitivity reason (1); Tracer logistics / cost limitation (1). **Total 6.**

Source map: S06 pp.20-29. **Closest past question:** 2024 Q1(b)ii-iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

Model 2 | Question 2: Digital subtraction angiography and CT geometry

2(a) [2 marks]

Question. For a ray through static tissue thickness xt and vessel thickness xv, write pre-contrast intensity I1. Use incident intensity I0 and attenuation coefficients μt and μb; assume monoenergetic radiation and negligible scatter.

Expected answer and working. The exponents add along the ray: I1=I0 exp[-(μt xt+μb xv)]. Here xt excludes the separately specified vessel path, so vessel thickness is not counted twice.

Proposed allocation: Sum of tissue and blood attenuation (1); Correct exponential transmission (1). **Total 2.**

Source map: S2 pp.27-28,48-49. **Closest past question:** 2023 Q2(a)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(b) [2 marks]

Question. After contrast injection, the vessel coefficient changes from μb to μc while other conditions are unchanged. Write I2.

Expected answer and working. I2=I0 exp[-(μt xt+μc xv)]. The post-contrast vessel coefficient replaces the blood coefficient in this defined simplified model; it is not an extra full blood layer.

Proposed allocation: Replace μb by μc on vessel path (1); Correct common tissue and I0 terms (1). **Total 2.**

Source map: S2 pp.27-28,48-49. **Closest past question:** 2023 Q2(a)ii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(c) [3 marks]

Question. Show why I2-I1 does not eliminate the static tissue signal.

Expected answer and working. I2-I1=I0 exp(-μt xt)[exp(-μc xv)-exp(-μb xv)]. The multiplicative factor exp(-μt xt) remains, so the difference still depends on tissue thickness and attenuation. Raw subtraction therefore does not isolate the vessel under this transmission model.

Proposed allocation: Factor common exponential (1); Correct bracket (1); Interpret remaining tissue dependence (1). **Total 3.**

Source map: S2 pp.27-28,48-49. **Closest past question:** 2023 Q2(a)iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(d) [4 marks]

Question. Show that ln(I2)-ln(I1) removes the static tissue term. Evaluate it for μc=1.8 cm^-1, μb=0.2 cm^-1 and xv=0.30 cm. State one acquisition condition required for cancellation.

Expected answer and working. lnI2-lnI1=[lnI0-μt xt-μc xv]-[lnI0-μt xt-μb xv]=-(μc-μb)xv. Thus the value is -(1.8-0.2)·0.30=-0.48, dimensionless. The negative sign follows the requested post-minus-pre convention; reversing it gives positive vessel contrast. Tissue registration/no motion and unchanged exposure normalization are required for cancellation.

Proposed allocation: Expansion of logs (1); Cancellation and signed result formula (1); -0.48 dimensionless (1); Matched motion/exposure condition (1). **Total 4.**

Source map: S2 pp.27-28,48-49. **Closest past question:** 2023 Q2(a)iv. **Status:** Recurring family.

Typical mark-losing error: Preserve the sign specified by the subtraction order.

2(e) [4 marks]

Question. An object has attenuation μ0=0.25 cm^-1 inside a disk of radius R=2.0 cm and zero outside. Derive its parallel-ray projection p(s,θ), where s is the signed perpendicular distance from the disk centre. Evaluate p at s=1.2 cm.

Expected answer and working. For |s|≤R the ray chord has half-length √(R²-s²). Since μ0 is constant, p(s,θ)=2μ0√(R²-s²), independent of θ. Outside the disk it is zero. At s=1.2 cm, chord length=2√(4-1.44)=3.2 cm, so p=0.25·3.2=0.80 (dimensionless). This is a line integral, not transmitted intensity; I/I0=exp(-p).

Proposed allocation: Chord geometry (1); Projection and outside case (1); Angle independence (1); p=0.80 (1). **Total 4.**

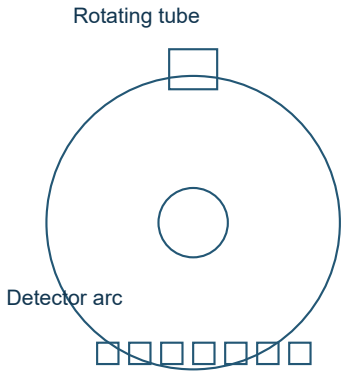
Source map: S3 pp.4,11-16. **Closest past question:** 2023 Q2(b)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

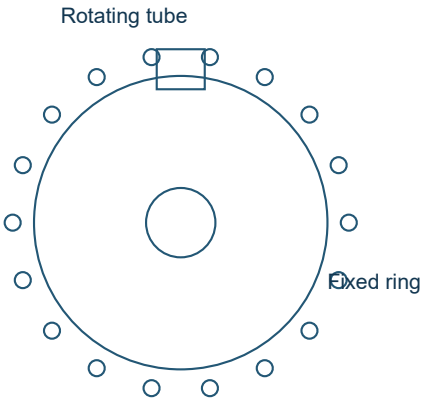
2(f) [6 marks]

Question. Compare third- and fourth-generation CT scanner geometry. Give three distinct features of each, supported by simple labeled sketches.

Expected answer and working. Third generation: the fan-beam X-ray tube and detector arc rotate together; the detector arc covers the fan beam; focused detector collimation is practical, while detector calibration drift can cause rings. Fourth generation: only the tube rotates; a stationary complete detector ring surrounds the patient; detectors can be referenced as the source moves but fixed focused collimation is difficult. Award three valid geometric/performance features per generation; the sketches must show what moves.



3rd: tube + detector arc rotate



4th: tube rotates; ring fixed

Proposed allocation: Third generation three features / diagram (3); Fourth generation three features / diagram (3). **Total 6.**

Source map: S3 pp.37-38,57-58. **Closest past question:** 2023 Q2(b)ii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(g) [4 marks]

Question. Name four routine CT quality-assurance tests and state what each tests.

Expected answer and working. CT-number accuracy checks known water/air or material HU; uniformity compares the same homogeneous phantom region across the field; spatial resolution assesses small separable structures or an MTF/line pattern; slice-thickness accuracy checks the reconstructed axial thickness against a phantom or prescribed thickness. Image noise is an alternative valid test if linked to a uniform region and repeatable protocol.

Proposed allocation: HU accuracy + purpose (1); Uniformity + purpose (1); Spatial resolution + purpose (1); Slice thickness or noise + purpose (1). **Total 4.**

Source map: S3 pp.31-32 for HU; no explicit QA lecture page supplied. **Closest past question:** 2023 Q2(b)iii. **Status:** Recurring family.

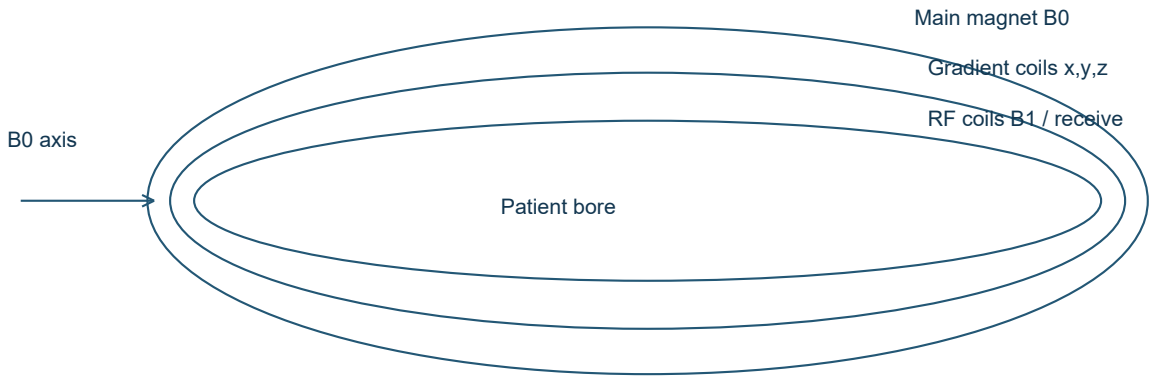
Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

Model 2 | Question 3: MRI hardware and phantom measurements

3(a) [6 marks]

Question. Draw the main components of an MRI scanner and explain the functions of the main magnet, three gradient-coil sets, and RF transmit/receive coils.

Expected answer and working. The main magnet establishes a strong approximately uniform static B0 field. Three independently controlled gradient-coil sets introduce spatial variation in the z-directed field along x, y and z for slice selection and spatial encoding. The RF transmit coil applies B1 at the resonance frequency to tip magnetization; receive coils detect voltage induced by precessing transverse magnetization. The coils surround the patient bore; RF coils lie nearer the subject than the main magnet. RF transmission and reception may use one switched coil or separate coils.



Proposed allocation: Magnet and B0 function (2); Three gradients and spatial encoding (2); RF excitation and induction reception (2). **Total 6.**

Source map: S4 pp.93-101. **Closest past question:** 2023 Q3(a)ii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(b) [3 marks]

Question. State three requirements of a main imaging magnet and explain why each matters.

Expected answer and working. High field strength supports spin polarization and signal; spatial homogeneity avoids unintended position-dependent frequency offsets and distortions; temporal stability avoids frequency drift and inconsistent encoding. Adequate patient access or controlled fringe field are also valid requirements if the physical reason is stated.

Proposed allocation: One requirement + reason (1); Second requirement + reason (1); Third requirement + reason (1). **Total 3.**

Source map: S4 p.94. **Closest past question:** 2023 Q3(a)iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(c) [4 marks]

Question. Using the provided high-temperature spin-population approximation, calculate the net proton population imbalance (Nparallel-Nantiparallel)/Ntotal as a percentage at B0=3.0 T and 27°C. Explain why the phantom volume is unnecessary.

Expected answer and working. Convert T=27+273.15=300.15 K. Fraction δ=γhB0/(4πkT)=(267.52e6·6.63e-34·3)/(4π·1.38e-23·300.15)=1.0222656e-5. Percentage=0.00102227%. This is the small excess parallel population over the antiparallel population, not the total fraction in the parallel state. Since Ntotal divides out, neither volume nor number of water molecules is needed for this fraction.

Proposed allocation: Kelvin conversion (1); Fraction equation and substitution (1); 0.001022% with correct scale (1); Population difference and volume cancellation (1). **Total 4.**

Source map: S4 pp.12-17. **Closest past question:** 2023 Q3(b)i. **Status:** Recurring family.

Typical mark-losing error: Multiply the population fraction by 100 for percent; it is not 0.1% or the entire parallel population.

3(d) [4 marks]

Question. A central 5.0 mm thick slice is selected with Gz=0.40 mT/cm at 3.0 T. Calculate the full RF bandwidth and absolute frequency endpoints.

Expected answer and working. Gz=0.40e-3/0.01=0.040 T/m. Full ΔB=GzΔz=0.040·0.0050=2.0e-4 T. With γbar=γ/(2π), Δf=8515.4261 Hz. f0=127.73139113 MHz and half-bandwidth=4257.7130 Hz. Endpoints are 127,727,133.41 and 127,735,648.84 Hz. The slice is centred at the magnet isocentre; no spatial offset correction is needed.

Proposed allocation: Gradient conversion (1); Full bandwidth relation (1); 8515 Hz (1); Correct endpoints (1). **Total 4.**

Source map: S4 pp.15,48-54. **Closest past question:** 2023 Q3(b)ii-iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(e) [4 marks]

Question. A conventional single-slice acquisition has 240 readout samples and 160 phase-encoding steps. TR=1.25 s and two averages are acquired. Calculate the scan time. If only readout samples are doubled, explain the effect on the ideal scan time under the same TR.

Expected answer and working. Tscan=Nphase·TR·NEX=160·1.25·2=400 s=6 min 40 s. The readout sample count is collected within each readout window, whereas the phase dimension requires repetitions. Under the explicit assumption that the doubled readout still fits within the same TR, doubling readout samples does not change the ideal 400 s scan time. It can alter bandwidth/readout duration and resolution, so the unchanged-TR assumption matters.

Proposed allocation: Nphase not Nread (1); Include two averages (1); 400 s / 6 min 40 s (1); Readout-vs-repetition explanation (1). **Total 4.**

Source map: S4 pp.61-62,71,92. **Closest past question:** 2025 Q3(b)iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(f) [4 marks]

Question. Give two phantom-based MRI checks and explain how an appropriate phantom feature would support each measurement.

Expected answer and working. A homogeneous water region supports B0 homogeneity/frequency uniformity measurements or image uniformity assessment under fixed coil conditions. A phantom with known thin ramps or geometry can assess slice thickness or geometric accuracy. Alternatively, known point/line spacings assess spatial resolution, and repeat images or a valid noise method assess SNR. A plain sphere by itself does not automatically provide all these calibrations; the feature and acquisition must match the metric.

Proposed allocation: First valid metric (1); Suitable phantom feature/method (1); Second valid metric (1); Suitable feature/method (1). **Total 4.**

Source map: S4 pp.52-54,92,94,98; phantom-check design is applied context. **Closest past question:** 2023 Q3(b)iv. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

Model 2 | Question 4: Ultrasound and thermal imaging

4(a) [3 marks]

Question. State three advantages and three limitations of ultrasound compared with CT or MRI.

Expected answer and working. Advantages: portability, real-time high temporal resolution, and no ionizing radiation. Limitations: operator dependence, restricted acoustic windows/field of view, and difficulty imaging through bone or air. Award 0.5 for each distinct correct item; cost is an acceptable alternative advantage.

Proposed allocation: Three advantages (1.5); Three limitations (1.5). **Total 3.**

Source map: S05 pp.2-5. **Closest past question:** 2023 Q4(a)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(b) [6 marks]

Question. Explain reflection, scattering and absorption of ultrasound in tissue. Link each to an image consequence.

Expected answer and working. Reflection occurs at impedance discontinuities and creates boundary echoes; a strong smooth interface can redirect energy away from the probe. Scattering from small/irregular structures distributes sound in many directions; returning components provide tissue texture and coherent interference generates speckle. Absorption converts acoustic energy to heat and reduces echo amplitude with path length, contributing to loss of deep signal and posterior shadowing behind highly attenuating structures.

Proposed allocation: Reflection mechanism and image effect (2); Scattering mechanism and image effect (2); Absorption mechanism and image effect (2). **Total 6.**

Source map: S05 pp.13-16,35-38. **Closest past question:** 2023 Q4(a)ii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(c) [5 marks]

Question. A target lies 18 cm deep in tissue with attenuation 0.70 dB/(cm·MHz). The receiver accepts at most 95 dB propagation loss. Choose the highest usable frequency from 2, 3 and 4 MHz. Justify using calculated losses and the resolution trade-off.

Expected answer and working. Round-trip loss is $L=2\alpha f d=2\cdot 0.70\cdot 18\cdot f=25.2f$ dB. For 2, 3 and 4 MHz, losses are 50.4, 75.6 and 100.8 dB. Frequencies 2 and 3 MHz are usable; choose 3 MHz for improved resolution at a fixed pulse-cycle count and focus geometry. Four MHz exceeds 95 dB, while choosing 2 MHz would unnecessarily sacrifice resolution under the stated assumptions.

Proposed allocation: Round trip in loss expression (1); All three losses (2); Select 3 MHz (1); Explain resolution/penetration balance (1). **Total 5.**

Source map: S05 pp.16,28-29. **Closest past question:** 2023 Q4(a)iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(d) [3 marks]

Question. At 18 cm imaging depth and c=1500 m/s, the required B-mode frame rate is 50 frames/s. Find the maximum integer number of sequential scan lines per frame, ignoring overhead.

Expected answer and working. A pulse echo needs tline=2·0.18/1500=0.000240 s. One frame is 1/50=0.020 s. Nlines≤0.020/0.000240=83.333, so at most 83 complete lines can be acquired. Round down: 84 lines would exceed the frame period.

Proposed allocation: 240 μs line time (1); Ratio gives 83.333 (1); 83 integer lines with reason (1). **Total 3.**

Source map: S05 pp.6-7,17,19. **Closest past question:** 2023 Q4(a)iv. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(e) [4 marks]

Question. With a labeled block diagram, explain how an uncooled thermal camera forms a medical thermogram.

Expected answer and working. Warm skin emits infrared radiation. IR-transmitting optics focus it onto a focal-plane detector array. In a microbolometer, absorbed radiation raises detector temperature and changes its electrical resistance. Readout electronics and an ADC digitize the signal; calibration/nonuniformity correction and emissivity assumptions support conversion to a displayed surface-temperature map. The map is not a direct image of deep anatomy.



Proposed allocation: Emitted IR and optics (1); Array/microbolometer conversion (1); Readout/ADC and calibrated processing (1); Surface-temperature display with labels (1). **Total 4.**

Source map: A1 PDF pp.3-5; syllabus p.1. **Closest past question:** 2023 Q4(b)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(f) [2 marks]

Question. State two appropriate medical uses of thermal imaging and link each to the physiological information observed.

Expected answer and working. Diabetic-foot follow-up can monitor local temperature asymmetry associated with inflammation or altered perfusion. Burn/wound monitoring can compare surface-temperature patterns related to perfusion and healing. Other justified uses include inflammatory-joint monitoring or controlled surface-temperature screening. A thermal pattern is nonspecific and requires clinical context.

Proposed allocation: First use + observed physiology (1); Second use + observed physiology (1). **Total 2.**

Source map: A1 PDF pp.5-7. **Closest past question:** 2023 Q4(b)ii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(g) [2 marks]

Question. Give two advantages of thermal imaging and two acquisition controls needed for reliable comparisons over time.

Expected answer and working. Advantages include noncontact measurement and no ionizing radiation, enabling repeated observations. Controls include stable ambient conditions/patient acclimatization and consistent camera calibration, emissivity, distance and view. Credit two distinct controls; changes in room conditions can mimic physiological change.

Proposed allocation: Two advantages (1); Two acquisition controls (1). **Total 2.**

Source map: A1 PDF pp.3,8-9. **Closest past question:** 2023 Q4(b)iii; controls are assignment coverage. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.